

Smart EMS — Employee Management System

Version: 1.0 | Framework: Laravel 13 | UI: Tailwind CSS + Blade Templates

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1. System Overview

Smart EMS is a web-based Employee Management System built with Laravel 13. It provides role-based dashboards for Admin, HR, and Employee users with full CRUD operations for employees, attendance, leave, notifications, payroll, and reports.

Key Roles:

- Admin** — Full system access, permission management
- HR** — Manage employees, attendance, leave, notifications, payroll, view reports
- Employee** — View own attendance, leave, notifications, dashboard

2. Technology Stack

Category	Technology
Backend Framework	Laravel 13 (PHP 8.3+)

Frontend CSS	Tailwind CSS (via CDN)
Templating	Blade Templates
Charts	Chart.js (via CDN)
Icons	Font Awesome 6
Database	MySQL
Authentication	Laravel Built-in Auth (session-based)
Admin Panel	Filament v5.6 (installed, not primary UI)
HTTP Middleware	Custom CheckRole + CheckPermission

3. Architecture

The system follows a standard Laravel MVC pattern with role-based and permission-based access control.

```

Smart_Ems_System/
├── app/
│   ├── Http/
│   │   ├── Controllers/
│   │   │   ├── Admin/PermissionController.php
│   │   │   ├── AttendanceLeave/AttendanceDashboardController.php
│   │   │   ├── AttendanceLeave/LeaveController.php
│   │   │   ├── Auth/{Login,Register,ForgotPassword,ResetPassword}.php
│   │   │   ├── DashboardController.php
│   │   │   ├── EmployeeController.php
│   │   │   ├── HomeController.php
│   │   │   ├── NotificationManagement/NotificationController.php
│   │   │   ├── PayrollController.php
│   │   │   └── ReportController.php
│   │   ├── Middleware/
│   │   │   ├── CheckRole.php
│   │   │   ├── CheckPermission.php
│   │   │   └── RedirectIfAuthenticated.php
│   │   └── Models/
│   │       ├── Attendance.php
│   │       ├── CompanySetting.php
│   │       ├── Employee.php
│   │       ├── Leave.php
│   │       ├── Notification.php
│   │       ├── Payroll.php
│   │       ├── Permission.php
│   │       └── User.php
│   └── Notifications/PasswordReset.php
├── database/
│   ├── migrations/ (14 migrations)
│   └── seeders/DatabaseSeeder.php
├── resources/
│   └── views/
│       ├── CoreSystem/layouts/ (app, base, nav, sidebar)
│       ├── CoreSystem/dashboard/ (admin, hr, employee, index)
│       ├── CoreSystem/auth/ (login, register, forgot-password, reset-password)
│       ├── Admin/permissions/index.blade.php
│       └── PayrollReport/payroll/ (index, create)

```

```

| | | AttendanceLeave/attendance/ (index, create, edit, report, etc.)
| | | EmployeeManagement/employees/ (index)
| | | employees/ (index, create, edit)
| | | NotificationManagement/notifications/ (index, create, edit, show, etc.)
| | | report.blade.php
| | | js/app.css
| | routes/
| | | web.php
| | | CoreSystem/core.routes.php
| | | PayrollReport/payroll.routes.php
| | | EmployeeManagement/employee.routes.php
| | | AttendanceLeave/attendance.php
| | bootstrap/app.php (middleware aliases)

```

4. Features

Module	Features	Roles
Dashboard	Stats cards, charts (attendance trend, department distribution, working hours), live data polling	Admin HR Employee
Employee Management	CRUD operations, employee profiles, department/position management	Admin HR
Attendance	Check-in/out, status tracking (Present/Late/Absent/Undertime), monthly reports, working hours	Admin HR Employee
Leave Management	Apply leave, approve/reject, leave tracking by type (Annual/Sick/Casual)	Admin HR Employee
Notifications	Create/manage notifications, categories, priorities, department targeting, pin/unpin	Admin HR Employee
Payroll	Generate payroll, mark as paid, bonus/deduction management, net salary calculation	Admin HR
Reports	Attendance reports, employee distribution by department	Admin
Permission Management	Define permissions, assign to roles (admin/hr/employee), granular access control	Admin

5. Database Schema

Users Table

Column	Type	Notes
id	BIGINT (auto)	Primary key
name	VARCHAR(255)	User full name

email	VARCHAR(255)	Unique
password	VARCHAR(255)	Hashed
role	VARCHAR(255)	admin hr employee
remember_token	VARCHAR(100)	Nullable
created_at / updated_at	TIMESTAMP	

Employees Table

Column	Type	Notes
id	BIGINT (auto)	Primary key
user_id	BIGINT (FK)	References users.id
employee_id	VARCHAR(255)	Unique employee code
name	VARCHAR(255)	
department	VARCHAR(255)	e.g., Engineering, Marketing
position	VARCHAR(255)	e.g., Software Developer
status	VARCHAR(255)	Active Inactive
email	VARCHAR(255)	
phone	VARCHAR(30)	
joined	DATE	Join date
address	TEXT	
image	VARCHAR(255)	Nullable, URL
present_days	INT	Default 0
leave_taken	INT	Default 0
salary	DECIMAL(10,2)	Monthly salary
projects	INT	Default 0

Attendances Table

Column	Type	Notes
id	BIGINT (auto)	Primary key

employee_id	BIGINT (FK)	References employees.id
status	VARCHAR(255)	Present Late Absent Undertime
date	DATE	
check_in	TIME	Nullable
check_out	TIME	Nullable
working_hours	DECIMAL(5,1)	Default 0.00

Leaves Table

Column	Type	Notes
id	BIGINT (auto)	Primary key
employee_id	BIGINT (FK)	
type	VARCHAR(255)	Annual Sick Casual
from_date	DATE	
to_date	DATE	
days	INT	
reason	TEXT	
status	VARCHAR(255)	Approved Pending Rejected
approver	VARCHAR(255)	Nullable

Notifications Table

Column	Type	Notes
id	BIGINT (auto)	Primary key
title	VARCHAR(255)	
description	TEXT	
category	VARCHAR(255)	General Holiday Meeting HR Policies Training Events Company
department	VARCHAR(255)	Nullable, for department targeting
priority	VARCHAR(255)	Low Medium High
is_pinned	BOOLEAN	Default false

published_by	VARCHAR(255)	
publish_date	DATETIME	
attachment	VARCHAR(255)	Nullable
status	BOOLEAN	Default true

Payrolls Table

Column	Type	Notes
id	BIGINT (auto)	Primary key
employee_id	BIGINT (FK)	
month	VARCHAR(255)	January - December
year	INT	
bonus	DECIMAL(10,2)	Default 0
unpaid_leave_days	INT	Default 0
leave_deduction	DECIMAL(10,2)	Calculated
net_salary	DECIMAL(10,2)	Calculated
payment_date	DATE	Nullable
status	VARCHAR(255)	Pending Paid

Company Settings Table

Column	Type	Notes
id	BIGINT (auto)	Primary key
monthly_working_hours	INT	Default 205
annual_leave_days	INT	Default 12
weekly_holiday	VARCHAR(255)	Default Saturday

Permissions Tables (New)

Table	Columns	Notes
permissions	id, name (unique), label, group, timestamps	Defines all system permissions (e.g., view_payroll, manage_employees)

<code>role_permissions</code>	id, role (string), permission_id (FK), timestamps	Maps permissions to roles (admin, hr, employee)
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6. Models

User Model ([app/Models/User.php](#))

Roles: admin, hr, employee

Key Methods:

Method	Return	Description
<code>isAdmin()</code>	bool	Checks if role is admin
<code>isHr()</code>	bool	Checks if role is hr
<code>isEmployee()</code>	bool	Checks if role is employee
<code>hasRole(string \$role)</code>	bool	Checks exact role match
<code>hasAnyRole(array \$roles)</code>	bool	Checks if user has any of the given roles
<code>hasPermission(string \$permission)</code>	bool	Checks if user has a specific permission (admins auto-pass)
<code>hasAnyPermission(array \$permissions)</code>	bool	Checks if user has any of the given permissions (admins auto-pass)
<code>employee()</code>	HasOne	Relationship to Employee model

Permission Model ([app/Models/Permission.php](#))

Fields: name, label, group

Relationship: `roles()` — BelongsToMany to User via `role_permissions` pivot table (pivots on the `role` column)

Other Models

Model	File	Key Relationships
Employee	<code>app/Models/Employee.php</code>	hasMany Attendance, belongsTo User
Attendance	<code>app/Models/Attendance.php</code>	belongsTo Employee
Leave	<code>app/Models/Leave.php</code>	belongsTo Employee
Notification	<code>app/Models/Notification.php</code>	No Eloquent relationships
Payroll	<code>app/Models/Payroll.php</code>	belongsTo Employee

CompanySetting	app/Models/CompanySetting.php	Singleton (id=1)
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7. Controllers

Controller	File	Key Methods
DashboardController	app/Http/Controllers/ DashboardController.php	index() — landing, admin() — admin dashboard with charts, hr() — HR dashboard, employee() — employee dashboard, chartData() / hrChartData() — JSON chart data
EmployeeController	app/Http/Controllers/ EmployeeController.php	index(), create(), store(), edit(), update(), destroy() — Full CRUD with validation
PayrollController	app/Http/Controllers/ PayrollController.php	index(), create(), store(), edit(), update(), markAsPaid(), destroy() — Payroll CRUD with salary calculation
PermissionController	app/Http/Controllers/Admin/ PermissionController.php	index(), store(), update(), destroy() — Permission CRUD and role assignment
AttendanceDashboardController	app/Http/Controllers/ AttendanceLeave/ AttendanceDashboardController.php	index(), create(), store(), edit(), update(), destroy(), report() — Attendance CRUD + reports
LeaveController	app/Http/Controllers/ AttendanceLeave/LeaveController.php	index(), store(), approve(), reject() — Leave management
NotificationController	app/Http/Controllers/ NotificationManagement/ NotificationController.php	index(), create(), store(), edit(), update(), destroy(), pin(), unpin(), show()
ReportController	app/Http/Controllers/ ReportController.php	index() — Attendance & distribution reports

HomeController	app/Http/Controllers/ HomeController.php	index() — Landing page
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8. Routes

Route Files

File	Prefix	Middleware	Description
routes/web.php	/	guest/auth	Login, register, logout, report
routes/CoreSystem/ core.routes.php	/core, /admin, / hr, /employee	web, auth, role/permission	Dashboards + permission management
routes/PayrollReport/ payroll.routes.php	/payroll	web, auth, permission:view_payroll	Payroll CRUD
routes/EmployeeManagement/ employee.routes.php	/employees	web, auth, permission:view_employees	Employee CRUD
routes/AttendanceLeave/ attendance.php	/attendance, / leave, /notifications	web, auth, permission:view_*	Attendance, Leave, Notification routes

Permission Middleware Mapping

Permission	Used In
view_dashboard	All dashboard routes (admin, hr, employee)
view_employees	/employees routes
manage_employees	/employees create/edit/delete routes
view_attendance	/attendance routes (view)
manage_attendance	/attendance create/edit/delete routes
view_leave	/leave index
manage_leave	/leave store/approve/reject
view_notifications	/notifications routes (view)
manage_notifications	/notifications create/edit/delete/pin
view_payroll	/payroll routes (view)

<code>manage_payroll</code>	/payroll create/edit/delete/mark-paid
<code>view_reports</code>	/report route
<code>manage_reports</code>	/report admin-only features

9. Views & Layouts

Layout Files

Layout	File	Description
<code>app.blade.php</code>	<code>resources/views/CoreSystem/layouts/app.blade.php</code>	Main layout with sidebar — used by all role-based pages (dashboard, employees, attendance, leave, notifications, payroll)
<code>base.blade.php</code>	<code>resources/views/CoreSystem/layouts/base.blade.php</code>	Legacy layout with sidebar (no longer used by payroll)
<code>sidebar.blade.php</code>	<code>resources/views/CoreSystem/layouts/sidebar.blade.php</code>	Sidebar navigation with permission-based link visibility
<code>nav.blade.php</code>	<code>resources/views/CoreSystem/layouts/nav.blade.php</code>	Top navbar with permission-based link visibility

View Files

View	File	Layout
Admin Dashboard	<code>resources/views/CoreSystem/dashboard/admin.blade.php</code>	<code>app.blade.php</code>
HR Dashboard	<code>resources/views/CoreSystem/dashboard/hr.blade.php</code>	<code>app.blade.php</code>
Employee Dashboard	<code>resources/views/CoreSystem/dashboard/employee.blade.php</code>	<code>app.blade.php</code>
Permission Management	<code>resources/views/Admin/permissions/index.blade.php</code>	<code>app.blade.php</code>
Payroll Dashboard	<code>resources/views/PayrollReport/payroll/index.blade.php</code>	<code>app.blade.php</code>
Payroll Create/Edit	<code>resources/views/PayrollReport/payroll/create.blade.php</code>	<code>app.blade.php</code>
Report	<code>resources/views/report.blade.php</code>	<code>app.blade.php</code>
Employees Index	<code>resources/views/employees/index.blade.php</code>	<code>app.blade.php</code>
Employees Create/Edit	<code>resources/views/employees/create.blade.php</code> , <code>edit.blade.php</code>	<code>app.blade.php</code>

Notifications	resources/views/NotificationManagement/notifications/ (8 files)	app.blade.php
Attendance	resources/views/AttendanceLeave/attendance/ (7 files)	app.blade.php

10. Middleware

CheckRole (app/Http/Middleware/CheckRole.php)

Registered as `role` alias. Checks if the authenticated user has one of the specified roles.

```
Route::middleware('role:admin')->group(...);
Route::middleware('role:admin,hr')->group(...);
```

CheckPermission (app/Http/Middleware/CheckPermission.php)

Registered as `permission` alias. Checks if the authenticated user has one of the specified permissions. Admins automatically pass all permission checks.

```
Route::middleware('permission:view_payroll')->group(...);
Route::middleware('permission:manage_employees')->group(...);
```

RedirectIfAuthenticated (app/Http/Middleware/RedirectIfAuthenticated.php)

Redirects authenticated users to their role-specific dashboard (admin → /admin/dashboard, hr → /hr/dashboard, employee → /employee/dashboard).

11. Permission System

How It Works

1. **Permissions are defined** in the `permissions` table with a name, label, and group.
2. **Permissions are assigned to roles** via the `role_permissions` pivot table.
3. **The User model** checks permissions by looking up the permission through the role relationship.
4. **Admins automatically have all permissions** — the `hasPermission()` method returns true for admins without checking the database.
5. **Route middleware** (`permission:xxx`) enforces access control at the route level.
6. **Sidebar and navbar** use `hasAnyPermission()` to conditionally show/hide navigation links.

Default Permissions (13 total)

Permission Name	Label	Group
view_dashboard	View Dashboard	general
manage_employees	Manage Employees	employees

view_employees	View Employees	employees
manage_attendance	Manage Attendance	attendance
view_attendance	View Attendance	attendance
manage_leave	Manage Leave	leave
view_leave	View Leave	leave
manage_notifications	Manage Notifications	notifications
view_notifications	View Notifications	notifications
manage_payroll	Manage Payroll	payroll
view_payroll	View Payroll	payroll
manage_reports	Manage Reports	reports
view_reports	View Reports	reports

Default Role-Permission Assignments

Role	Permissions Count	Permissions
Admin	13 (all)	All permissions
HR	11	All except manage_employees, manage_reports
Employee	7	view_dashboard, view_employees, view_attendance, manage_attendance, view_leave, manage_leave, view_notifications

Admin Permission Management UI

Admins can access the permission management page at `/admin/permissions` to:

- Add new permissions (name, label, group)
- Assign/unassign permissions to HR and Employee roles via checkboxes
- Delete existing permissions
- View all permissions in a table

12. Authentication & Authorization

Login Flow

1. User submits credentials at `/login`
2. `LoginController` authenticates the user
3. `RedirectIfAuthenticated` middleware redirects to role-specific dashboard
4. Session-based authentication maintained via Laravel's built-in session guard

Authorization Flow

1. **Route level:** `role:admin` or `permission:view_payroll` middleware checks access
2. **View level:** Blade directives like `@if(Auth::user()->hasPermission('manage_payroll'))` conditionally render UI elements
3. **Sidebar/Nav:** Navigation links are shown/hidden based on user permissions

13. Installation & Setup

Prerequisites

- PHP 8.3+
- MySQL 8.0+
- Composer
- Node.js (for Tailwind CSS)

Steps

```
# 1. Clone and install dependencies
composer install
npm install

# 2. Configure environment
cp .env.example .env
# Edit .env with database credentials

# 3. Generate application key
php artisan key:generate

# 4. Run migrations
php artisan migrate

# 5. Seed database (creates admin, hr, employee users + test data + permissions)
php artisan db:seed

# 6. Build assets
npm run build

# 7. Start development server
php artisan serve
```

Default Login Credentials

Role	Email	Password
Admin	admin@example.com	password
HR	hr@example.com	password
Employee	employee@example.com	password

14. Database Seeding

The `DatabaseSeeder` creates:

- **14 users** — 1 admin, 1 HR, 12 employees
- **13 employee records** linked to users
- **Attendance records** for the last 6 months (Saturdays off, weighted random statuses)
- **Leave records** — 2-4 per employee with mixed statuses
- **Notification records** — 8 sample notifications
- **Company settings** — monthly working hours (205), annual leave (12 days), weekly holiday (Saturday)
- **13 permissions** with role assignments (admin: all, hr: 11, employee: 7)

Note: The permission system is fully integrated. After seeding, admins can customize permissions at `/admin/permissions` to define fine-grained access control for HR and Employee roles.